Truth as a Compression Artifact
in Language Model Training

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Poster Session

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Measured against *everything* the error generator can produce, the preference for the correct answer is small, late, and not about truth.

The pairwise metric overstates it three- to fourfold; a coherent false rule is never filtered; a false rule the model can *learn* is preferred over the correct one.

Random errors, 86M
$59.0 \pm 0.8\%$
mass on the correct completion; null 50; nothing below 86M

One coherent rule
$\approx 50\%$
at every size, architecture and corpus: not filtered

Learnable false rule
43%
below the null: the *false* system is preferred, 418M–3.08B

1. The Question

Real corpora contain contradictions: the same problem appears with a correct and an incorrect solution. Which one does a transformer trained from scratch put its probability on, when frequency, format and domain are held equal? Minimizing cross-entropy minimizes code length (Shannon 1948; Delétang et al. 2024), which suggests an error the model cannot predict from the problem is harder to accommodate than one it can. **We use this as motivation only**; both experiments built to make it a measured mechanism turned out to measure something else.

2. Measure the Whole False Side

The usual metric is pairwise: does the correct completion beat *one* incorrect one? If the error can take N forms, a model that merely reproduces training frequencies puts 0.5 on the correct string and $0.5/N$ on each false one, and wins every pairwise comparison for $N \geq 2$.

We sum over every string the generator can produce instead:

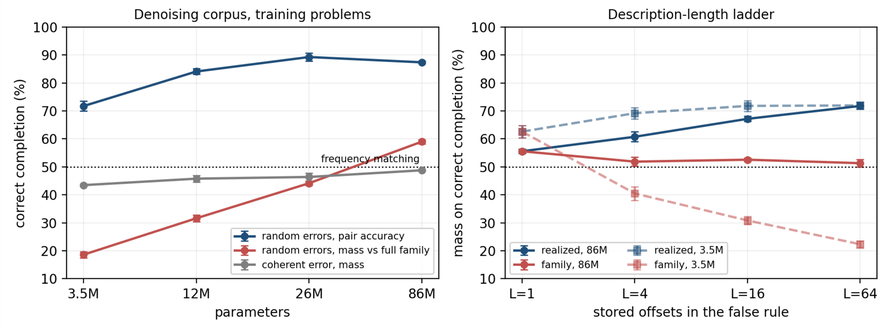
$$\text{share} = \frac{P(\text{correct})}{P(\text{correct}) + \sum_i P(\text{false}_i)}$$

Each problem occurs exactly twice in the primary corpus (once correct, once corrupted) and every other member of the family occurs zero times, so a frequency-matching model scores **exactly** 0.500 for any N. Adding strings can only lower the share: a value above 0.5 is a lower bound. Full-string log-probability, not a per-token mean; a length-matched subset removes the remaining channel. Pair accuracy is reported only for comparability.

3. Experimental Design

GPT-2-like decoder-only transformers trained *from scratch* (3.5M–86M, 4 seeds); Qwen3-architecture models at 418M–3.08B (2–4 seeds). AdamW, lr = $3 \cdot 10^{-4}$, 5,000 steps, character tokenizer. Four problem families (multi-step arithmetic, factoring, equation solving, differentiation) rendered as step-by-step derivations and verified with SymPy; only the *structure of the errors* varies: random per problem, one coherent false rule, N competing rules, a rule with L stored constants, or a false family whose selector the model can or cannot learn. Transfer: Wikipedia paragraphs with random or coherent entity substitution, and a 490M model on 1 GB of FineWeb-Edu with 8 % contradictory math. More than 500 training runs; per-run outputs released.

4. Central Contrast, and What the Pairwise Metric Adds



Left: denoising corpus, training problems, 4 seeds. Pair accuracy reads 72–89% at every size. Summing the generator's full support leaves an effect only at 86M: $59.0 \pm 0.8\%$ (every per-seed bootstrap interval above 0.5; 59.4% on an independent corpus). At 86M the realized-error score adds 27.6 pp of excess above chance and pair accuracy 37.4 pp where the family share gives 9.0 pp. The coherent rule sits at 43.5–48.8%; the shortfall is completion length (-0.58 tokens), not a preference.

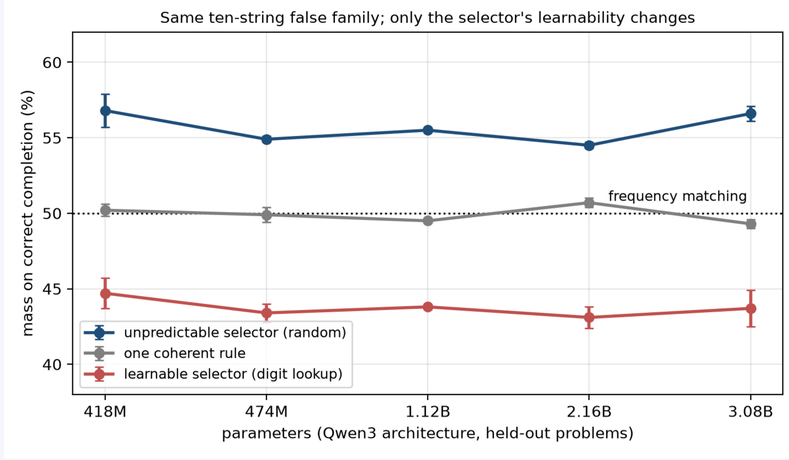
Right: the description-length ladder (a single false rule storing L constants). Against the one realized false string the share rises with L; against the full family it is flat at 86M (55.6, 51.9, 52.6, 51.3). **No grading in description length.**

5. Two Results We Withdraw

Multi-rule. With N competing false rules, pair accuracy jumps 47 → 79% at N=1 → 2 and reaches 90% at N=10. On the same checkpoints the aggregate share reads 47.9 → 47.8 → 38.7: the jump is *mass splitting*, not a selector cost. Reported as a calibration of how much pairwise signal splitting alone can manufacture.

Description-length ladder. Built to show a graded effect; the family measure is flat. A clean corpus with no false rule reads 99.7% at 86M and 64.1% at 3.5M on the same support, so the measure is near-unbiased where we report an effect and conservative where we report none. Compression remains an *interpretation* of the surviving contrast, not a quantity these experiments measure.

6. What Moves the Preference: Learnability



Held-out grid, no problem seen twice, family share on the length-matched subset, null 50. The false answer is $a \cdot b + T[s(p) \bmod 10]$ for one 10-entry table; only the selector $s(p)$ changes. Unpredictable selector: the correct system holds +5 points. Selector the model can learn (a visible digit): the correct share falls to $43.4 \pm 0.6\%$ at 474M (4 seeds), and the model prefers the *false* system. Partly learnable selectors (sums of two or three digits) sit at the null. Random and learnable share identical correct derivations, so competence is matched and the selector is the only variable. The contrast holds at 1.12B, 2.16B and 3.08B; a clean corpus reads 96–99% at every size.

7. Transfer and Noise

Wikipedia entity substitution (3.5M–86M, 4 seeds, pair accuracy): a coherent global mapping ("France"→"Japan") reads 46–49% against a 50% null; a consistent substitution is not filtered on real text either. The random column (70–71%) has an unenumerable support and a null near 100%, and is not interpreted. **490M on FineWeb-Edu** with 8 % contradictory math: coherent $47.4 \pm 0.7\%$, not filtered (2 seeds).

Noise ratio on the controlled measure (86M, 2 seeds): with K random errors per problem the null is $1/(K{+}1)$; the share reads 56.3, 44.9, 36.1 at K=1, 2, 4 against 50.0, 33.3, 20.0. The excess above the null does not shrink with noise.

8. Takeaway

what this does and does not support

- ► **The objective favors a learnable, internally consistent system, not a true one.** A coherent false rule keeps its corpus share at every size we tested; a false rule the model can learn is preferred over the correct one.
- ► **Pair accuracy is not a controlled diagnostic of preference.** Any design that varies the number of false forms (competing rules, noise ratios, entity substitution, corruption load) varies the metric with the treatment.
- ► **Not claimed:** compression as a measured mechanism, a graded description-length effect, an effect at every scale, or anything about production models.
- ► **For data curation:** diverse errors lose a little mass to the correct system; coordinated errors with a learnable rule are exactly the kind this preference favors.